

Conditional Coverage

Training

We first examine the coverage of split conformal prediction conditional on the training set:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n).$$

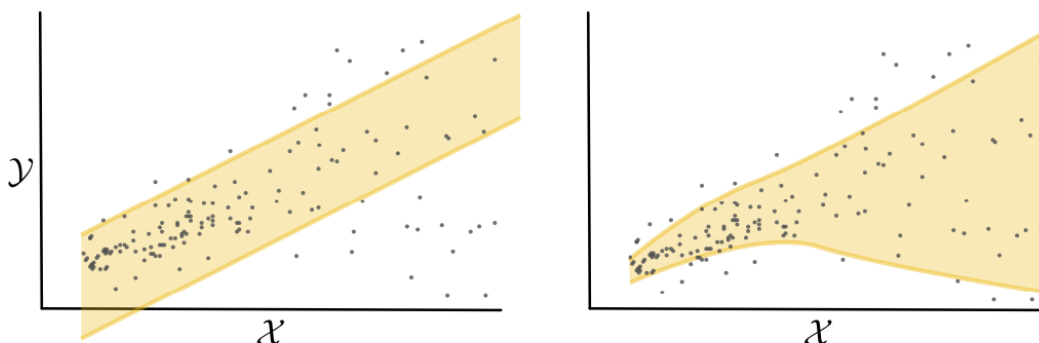
This is the empirical coverage that we would obtain if we used the points $\mathcal{D}_n = ((X_1, Y_1), \dots, (X_n, Y_n))$ for calibration, and then evaluated our coverage over an infinite test set of i.i.d. data. This quantity is a random variable, since it depends on the calibration set $\mathcal{D}_n$.

If the conformal prediction set satisfies marginal coverage at level $1 - \alpha$, then the training-conditional coverage must be $\geq 1 - \alpha$ in expectation:

$$\mathbb{E}[\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n)] = \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha.$$

However, a marginal coverage guarantee does not ensure that training-conditional coverage will be close to $1 - \alpha$ *with high probability*; for this, we need a stronger concentration-type statement.

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In fact, there is a closed-form expression that characterizes the training-conditional coverage in this setting. For intuition, let us first consider a simplified setting where the score distribution is continuous (so that there are no ties, almost surely), and where $(1 - \alpha)(n + 1)$ is an integer. In that case, training-conditional coverage is exactly distributed as a Beta distribution:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n) \sim \text{Beta}((1 - \alpha)(n + 1), \alpha(n + 1)).$$

$\geq 1 - \alpha$

The Beta distribution arises from the order statistics of the uniform distribution—conformal prediction relies only on the ranks of the scores, so if there are no ties one can without loss of generality analyze conformal scores that are uniformly distributed.

$$\mathbb{P}(Y_{n+1} \notin \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n) \leq \alpha$$

Theorem 4.1: Training-conditional coverage for split conformal prediction

Suppose the data points (X_i, Y_i) are i.i.d., and let $\mathcal{C}(X_{n+1})$ be constructed via split conformal prediction (Algorithm 3.6) using any pretrained score function s . Then the training-conditional coverage $\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n)$ stochastically dominates the Beta $((1 - \alpha)(n + 1), \alpha(n + 1))$ distribution, and in particular, for any $\Delta \geq 0$,

$$\mathbb{P}\left(\underbrace{\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n)}_{\substack{\downarrow \\ \mathbb{P}(\mathcal{C}(\mathcal{D}_n) \leq 1 - \alpha - \Delta)}} \leq 1 - \alpha - \Delta\right) \leq F_{\text{Beta}((1-\alpha)(n+1), \alpha(n+1))}(1 - \alpha - \Delta) \leq e^{-2n\Delta^2},$$

where $F_{\text{Beta}(a,b)}$ denotes the CDF of the Beta(a, b) distribution.

why

$$\mathbb{P}(\mathcal{C}(\mathcal{D}_n) \leq 1 - \alpha - \Delta)$$

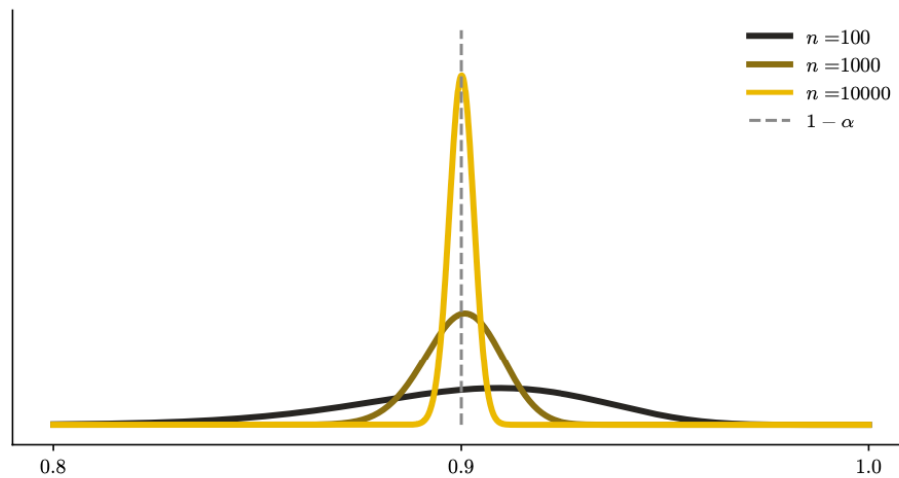


Figure 4.2: The distribution of coverage of split conformal prediction conditional on the training data, when the scores have no ties. The distribution concentrates around $1 - \alpha$ with rate $n^{-1/2}$.

Proof of Theorem 4.1

Throughout this proof, we will treat the pretrained score function as fixed—formally, recalling Algorithm 3.6, we are conditioning on the pretraining set $\mathcal{D}_{\text{pre}}$.

Let F be the CDF of the distribution of $s(X, Y)$, when the data point is sampled as $(X, Y) \sim P$. Define $S_i = s(X_i, Y_i)$ for $i \in [n+1]$, which are i.i.d. draws from the distribution with CDF F . Next, let $S_{(1)} \leq \dots \leq S_{(n)}$ be the order statistics of $S_1, \dots, S_n$. By definition of split conformal, we have $Y_{n+1} \in \mathcal{C}(X_{n+1})$ if and only if $S_{n+1} \leq S_{(k)}$ for $k = \lceil (1 - \alpha)(n + 1) \rceil$, and so we have

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n) = \mathbb{P}(S_{n+1} \leq S_{(k)} \mid \mathcal{D}_n) = F(S_{(k)}),$$

where the last step holds since S_{n+1} is independent of $\mathcal{D}_n$ and has CDF F , while $S_{(k)}$ is a function of $\mathcal{D}_n$. Therefore,

$$\mathbb{P}(\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n) \leq 1 - \alpha - \Delta) = \mathbb{P}(F(S_{(k)}) \leq 1 - \alpha - \Delta).$$

Next let $U_i = F(S_i)$ for all $i \in [n]$. Then the U_i 's are i.i.d., and by the basic property of CDFs (2.7), their shared distribution is superuniform—that is, $\mathbb{P}(U_i \leq \tau) \leq \tau$ for all $\tau \in [0, 1]$. Note also that, by monotonicity of F , the order statistics $U_{(1)} \leq \dots \leq U_{(n)}$ of $U_1, \dots, U_n$ satisfy $U_{(k)} = F(S_{(k)})$ (see Fact 2.13(i)). Then

$$\mathbb{P}(F(S_{(k)}) \leq 1 - \alpha - \Delta) = \mathbb{P}(U_{(k)} \leq 1 - \alpha - \Delta) \leq \mathbb{P}(U_{(k)}^* \leq 1 - \alpha - \Delta),$$

where $U_1^*, \dots, U_n^*$ are i.i.d. $\text{Unif}[0, 1]$ random variables, with order statistics $U_{(1)}^* \leq \dots \leq U_{(n)}^*$. Finally, by definition of the Beta distribution, the k th order statistic from n i.i.d. uniform random variables has distribution

$$U_{(k)}^* \sim \text{Beta}(k, n + 1 - k).$$

This distribution stochastically dominates the $\text{Beta}((1 - \alpha)(n + 1), \alpha(n + 1))$ distribution, since $k \geq (1 - \alpha)(n + 1)$. This proves the first inequality claimed in the theorem.

The second inequality (i.e., the bound on the CDF of the Beta distribution) is due to the fact that the $\text{Beta}((1 - \alpha)(n + 1), \alpha(n + 1))$ distribution has mean $1 - \alpha$, and is $\frac{1}{4n}$ -subgaussian.

We further look at this

$$\mathbb{P}\left(\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n) \leq 1 - \alpha - \Delta\right) \leq F_{\text{Beta}((1-\alpha)(n+1), \alpha(n+1))}(1 - \alpha - \Delta) \leq e^{-2n\Delta^2},$$

Treat the pretrained score function s as fixed, and define

$$c(\mathcal{D}_n) = \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid \mathcal{D}_n).$$

Marginal coverage averages over the calibration samples. By the law of total expectation,

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) = \mathbb{E}_{\mathcal{D}_n}[c(\mathcal{D}_n)].$$

Consequently,

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha \iff \mathbb{E}_{\mathcal{D}_n}[1 - c(\mathcal{D}_n)] \leq \alpha.$$

This controls the **average** miscoverage. It does not say how often a calibration sample produces coverage below $1 - \alpha$.

Equation (4.1) instead asks for

$$\mathbb{P}_{\mathcal{D}_n}(c(\mathcal{D}_n) \geq 1 - \alpha) \geq 1 - \delta.$$

We showed that

$$c(\mathcal{D}_n) \sim \text{Beta}(k, n + 1 - k).$$

for $k = \lceil (1 - \alpha)(n + 1) \rceil$,

When $(1 - \alpha)(n + 1)$ is an integer, this becomes exactly the book's formula:

$$c(\mathcal{D}_n) \sim \text{Beta}((1 - \alpha)(n + 1), \alpha(n + 1)).$$

Its mean and variance are

$$\mathbb{E}[c(\mathcal{D}_n)] = 1 - \alpha, \quad \text{Var}(c(\mathcal{D}_n)) = \frac{\alpha(1 - \alpha)}{n + 2}.$$

$$t = 1 - \alpha - \Delta$$

The theorem also handles ties and rounding. Saying that coverage "stochastically dominates" the displayed Beta distribution means that, for every threshold t ,

$$\mathbb{P}_{\mathcal{D}_n}(c(\mathcal{D}_n) \leq t) \leq \mathbb{P}(B \leq t),$$

where B has that Beta distribution. Therefore, its lower tail provides a bound on the chance of poor coverage.

In particular,

$$\mathbb{P}_{\mathcal{D}_n}(c(\mathcal{D}_n) \leq 1 - \alpha - \Delta) \leq e^{-2n\Delta^2}.$$

Let's look at why this is the case:

Write

$$k = \lceil (1 - \alpha)(n + 1) \rceil, \quad t = 1 - \alpha - \Delta.$$

First assume the score distribution is continuous. As shown earlier,

$$c(\mathcal{D}_n) = U_{(k)}, \quad U_1, \dots, U_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, 1).$$

Thus we need to bound $\mathbb{P}(U_{(k)} \leq t)$.

$$\mathbb{P}(c(\mathcal{D}_n) \leq t) = \mathbb{P}(U_{(k)} \leq t) = \mathbb{P}(B \geq k)$$

The k th smallest uniform is at or below t exactly when at least k uniforms are at or below t . Therefore,

$$\{U_{(k)} \leq t\} = \left\{ \sum_{i=1}^n \mathbf{1}\{U_i \leq t\} \geq k \right\}.$$

Let

$$B \sim \text{Bin}(n, t)$$

$$B = \sum_{i=1}^n \mathbf{1}\{U_i \leq t\}.$$

Each indicator is an independent Bernoulli variable with success probability t , so

$$B \sim \text{Binomial}(n, t), \quad \mathbb{E}[B/n] = t.$$

$$\lceil (1-\alpha)(n+1) \rceil$$

By the definition of k ,

$$\frac{k}{n} \geq \frac{(1-\alpha)(n+1)}{n} \geq 1-\alpha.$$

Consequently, if $B \geq k$, then

$$\frac{B}{n} - t \geq (1-\alpha) - t = \Delta.$$

Hence

$$\mathbb{P}(c(\mathcal{D}_n) \leq t) = \mathbb{P}(B \geq k) \leq \mathbb{P}\left(\frac{B}{n} - t \geq \Delta\right).$$

$\mathbb{E}\left[\frac{B}{n}\right]$

For an average of n independent Bernoulli variables, Hoeffding's inequality gives

$$\mathbb{P}\left(\frac{B}{n} - \mathbb{E}[B/n] \geq \Delta\right) \leq e^{-2n\Delta^2}.$$

Since $\mathbb{E}[B/n] = t$, combining the steps yields

$$\mathbb{P}_{\mathcal{D}_n}(c(\mathcal{D}_n) \leq 1-\alpha-\Delta) \leq e^{-2n\Delta^2}.$$

$$Z_i = \mathbb{1}\{u_i \leq t\}$$

$$\mathbb{P}\left(\frac{\sum Z_i}{n} - \mathbb{E}(Z) \geq \Delta\right) \leq e^{-2n\Delta^2}$$

$$\mathbb{E}[\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) | \mathcal{D}_n)] \geq 1 - \alpha \iff \mathbb{E}[1 - \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}) | \mathcal{D}_n)]$$

4.1.1 High-probability coverage

As described above, the marginal coverage property is equivalent to a bound on the *expected* training-conditional miscoverage,

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha \iff \mathbb{E}[\mathbb{P}(Y_{n+1} \notin \mathcal{C}(X_{n+1}) | \mathcal{D}_n)] \leq \alpha,$$

where the expectation is over the calibration data $\mathcal{D}_n$. We can also consider a different approach: instead of bounding the expected error, we might wish to bound the error with some desired probability—that is, in this setting, we enforce a bound on the chance that we end up with poor training-conditional coverage. Concretely, for some $\delta \in (0, 1)$, we could ask for a prediction set $\mathcal{C}(X_{n+1})$ that satisfies

$$\mathbb{P}(\mathbb{P}(Y_{n+1} \notin \mathcal{C}(X_{n+1}) | \mathcal{D}_n) \leq \alpha) \geq 1 - \delta. \quad (4.1)$$

In other words, there is only a δ chance that we draw a calibration set $\mathcal{D}_n$ leading to a miscoverage rate on future data that is higher than α .

In the special case of i.i.d. data, we can produce sets $\mathcal{C}(X_{n+1})$ that satisfy the error-control property in (4.1), with a slight modification of split conformal prediction. Taking as a starting point the distribution of training-conditional coverage given in Theorem 4.1, we see that by defining

$$\mathcal{C}(X_{n+1}) = \{y \in \mathcal{Y} : s(X_{n+1}, y) \leq \text{Quantile}(S_1, \dots, S_n; (1 - \alpha')(1 + 1/n))\}$$

(i.e., split conformal run at a nominal level $1 - \alpha'$), then we will achieve the desired high probability bound (4.1) if we choose the appropriate nominal level $1 - \alpha'$. In particular, given a fixed $\delta > 0$, we can take α' to be the unique value that satisfies

$$F_{\text{Beta}((1-\alpha')(n+1), \alpha'(n+1))}(1 - \alpha) = \delta.$$

By our calculations on the mean and variance of the Beta distribution, above, we can see that $\alpha' = \alpha - \mathcal{O}(n^{-1/2})$ —that is, this is simply a slightly more conservative version of split conformal prediction.

$\alpha' < \alpha \Rightarrow \text{why?}$

$$\mathbb{P}(1 - c(\mathcal{D}_n) \leq \alpha)$$

$$= \mathbb{P}(c(\mathcal{D}_n) \geq 1 - \alpha) \geq 1 - \delta$$

$$k = \lceil (1 - \alpha)(n + 1) \rceil$$

The corresponding Beta bound is

$$B_{\alpha'} \sim \text{Beta}((1 - \alpha')(n + 1), \alpha'(n + 1)).$$

An inadequate calibration sample has $c(\mathcal{D}_n) < 1 - \alpha$. Hence

$$\mathbb{P}_{\mathcal{D}_n}(c(\mathcal{D}_n) < 1 - \alpha) \leq F_{B_{\alpha'}}(1 - \alpha).$$

Choosing α' so that

$$F_{B_{\alpha'}}(1 - \alpha) = \delta$$

therefore gives the desired guarantee. The equation places the target coverage $1 - \alpha$ at the lower δ -quantile of the Beta bound.

$$\mathbb{P}(C(D_n) \geq 1 - \alpha) = \mathbb{P}(S(X_{n+1}, y) \leq \mathcal{I}(S_1, \dots, S_n, (1 - \alpha)(1 + \frac{1}{n})))$$

$$= F_{\text{Beta}}((1 - \alpha')(n+1), \alpha'(n+1))$$

$$= \delta$$

Why $O(n^{-1/2})$?

$$\boxed{\mathbb{P}_{\mathcal{D}_n}(c(\mathcal{D}_n) \leq 1 - \alpha - \Delta) \leq e^{-2n\Delta^2}} = e^{-\log(1/\delta)} = \delta$$

The $n^{-1/2}$ adjustment reflects the scale of the coverage fluctuations. More explicitly, apply the theorem with nominal error α' and

$$\Delta = \alpha - \alpha'.$$

Then

$$\mathbb{P}_{\mathcal{D}_n}(c(\mathcal{D}_n) < 1 - \alpha) \leq e^{-2n(\alpha - \alpha')^2}.$$

A simple sufficient choice, when positive, is

$$\alpha' = \alpha - \sqrt{\frac{\log(1/\delta)}{2n}}.$$

$$\delta = \alpha' - \alpha = \sqrt{\frac{\log(1/\delta)}{2n}}$$

$$O(n^{-1/2})$$